

Billar-Metegol

Simulación dirigida por eventos y búsqueda de la configuración óptima

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Introducción

Modelo matemático: partículas

- Vuelo libre:

$$x_i(t) = x_i(0) + v_{x_i} t, \quad y_i(t) = y_i(0) + v_{y_i} t$$

- Colisión elástica i - j ($\sigma = R_i + R_j$; $\Delta \mathbf{v} = \mathbf{v}_j - \mathbf{v}_i$, $\Delta \mathbf{r} = \mathbf{r}_j - \mathbf{r}_i$):

$$J = \frac{2 m_i m_j (\Delta \mathbf{v} \cdot \Delta \mathbf{r})}{\sigma (m_i + m_j)}$$

- Actualización post-choque:

$$vx_i^d = vx_i^a + J_x / m_i,$$

$$vy_i^d = vy_i^a + J_y / m_i,$$

$$vx_j^d = vx_j^a - J_x / m_j,$$

$$vy_j^d = vy_j^a - J_y / m_j.$$

Modelo matemático: contorno

- Obstáculo circular fijo ($m \rightarrow \infty$), ($c_n = c_t = 1$):

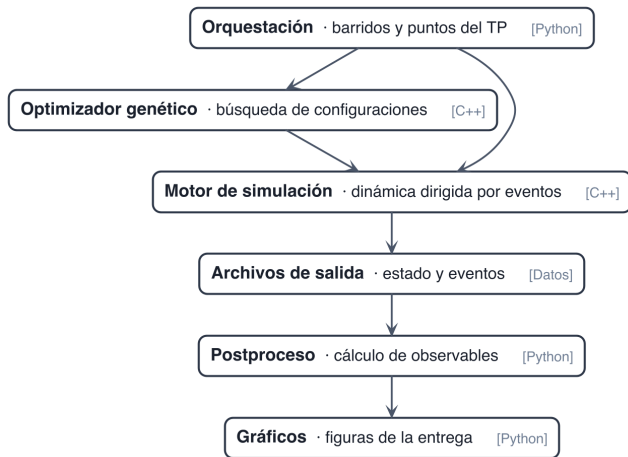
$$\hat{\mathbf{e}}_n = (\cos \alpha, \sin \alpha), \quad \hat{\mathbf{e}}_t = (-\sin \alpha, \cos \alpha)$$

$$\mathbf{v}^d = \mathbf{R}(-\alpha) \mathbf{S}(c_n, c_t) \mathbf{R}(\alpha) \mathbf{v}^a$$

$$\mathbf{R}(\alpha) = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}, \quad \mathbf{S}(c_n, c_t) = \begin{pmatrix} -c_n & 0 \\ 0 & c_t \end{pmatrix}$$

Implementación

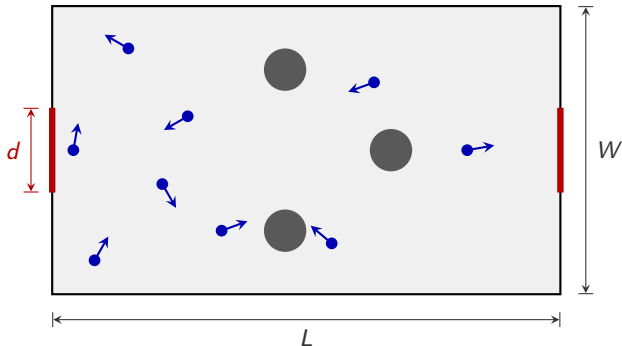
Arquitectura del código



Simulaciones

Geometría del sistema

- ▶ $L = 1.20$ m
- ▶ $W = 0.68$ m
- ▶ $d = 0.20$ m
- ▶ $r = 0.0175$ m
- ▶ $m = 0.025$ kg
- ▶ $v_0 = 1$ m/s
- ▶ $K \geq 1$ obstáculos



Observables

- ▶ $N_g(t)$: partículas que impactaron con un arco.
- ▶ $F_u(t) = N_g(t)/N$
- ▶ t_{90} : primer instante con $F_u(t) \geq 0.9$.
- ▶ $\langle t_{90} \rangle_n$: promedio de t_{90} sobre n realizaciones.
- ▶ Desplazamiento cuadrático medio ($d = 2$):

$$\text{DCM}(t) = \frac{1}{N} \sum_{i=1}^N |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \approx 2 d D t = 4 D t$$

- ▶ Coeficiente de difusión (D):

$$D = \frac{c^*}{4}$$

con c^* la pendiente óptima.

Configuraciones estudiadas

Acá van imágenes.

Algoritmo genético

Fitness
$$f(s) = \begin{cases} t_{90}(s) & F_u \geq 0.9 \\ t_{\text{máx}} + \alpha (90 - N_g(t_{\text{máx}})) & \text{si no} \end{cases}$$

$$\langle t_{90} \rangle_5 = \frac{1}{5} \sum_{j=1}^5 f(s_j),$$

Selección Torneo de 3

Cruza BLX- α : $g_h \sim U[g_{\text{mín}} - 0.3 \Delta, g_{\text{máx}} + 0.3 \Delta]$, $\Delta = g_{\text{máx}} - g_{\text{mín}}$

Mutación $g' = g + \mathcal{N}(0, \sigma^2)$ con prob. 0.3 por gen, σ decrece con las generaciones

Elitismo 2 mejores

Resultados

Animaciones

Mesa vacía

Fotograma del video

Video: <https://youtu.be/TODO>

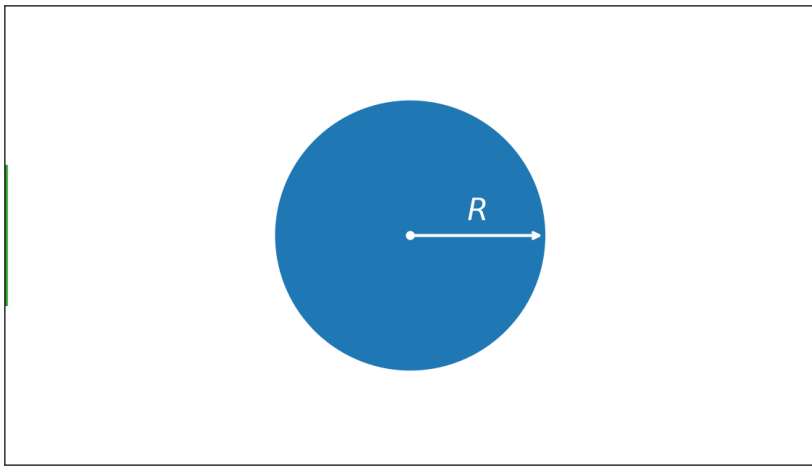
Configuración óptima

Fotograma del video

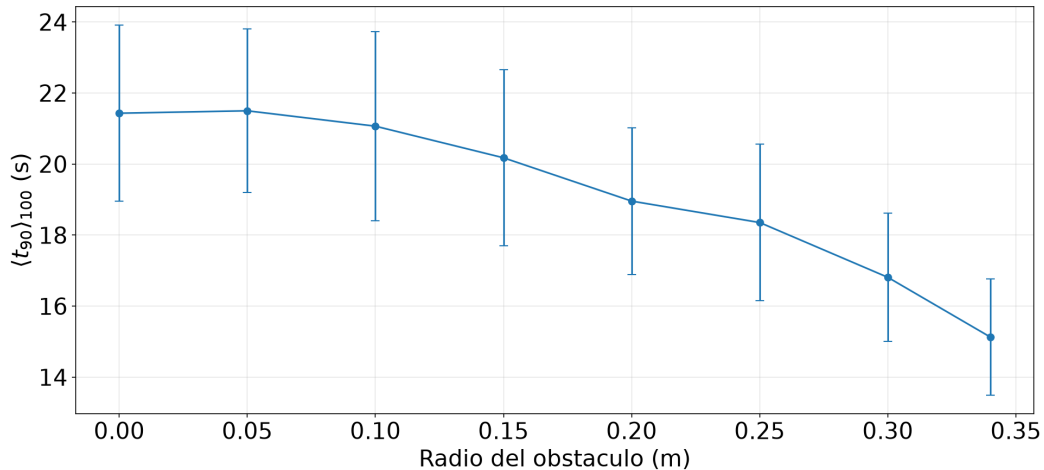
Video: <https://youtu.be/TODO>

Círculo: codificación

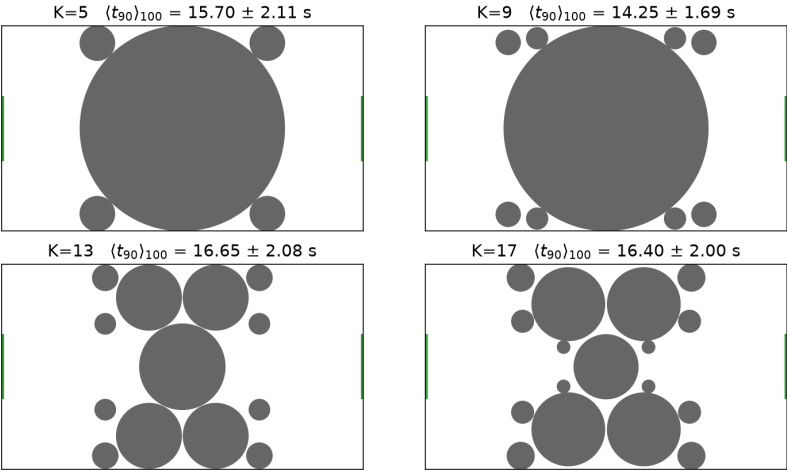
$$R \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.34\} \text{ m}$$



$\langle t_{90} \rangle_{100}$ vs Radio del círculo

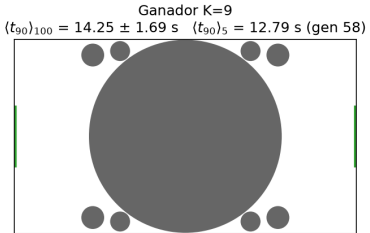
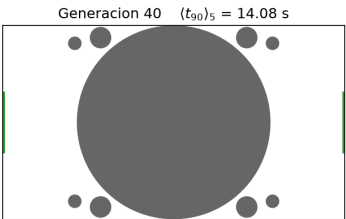
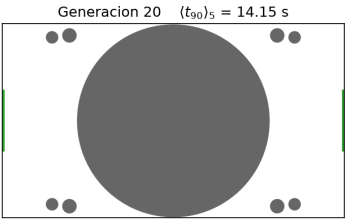
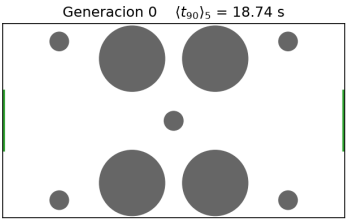


Quad: geometrías propuestas por el GA

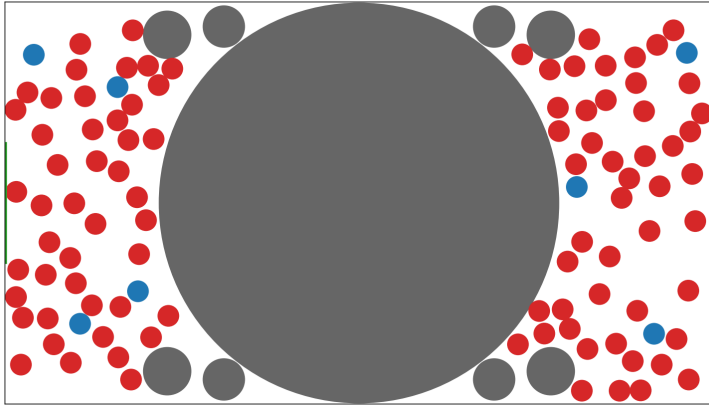


Quad: evolución del ganador

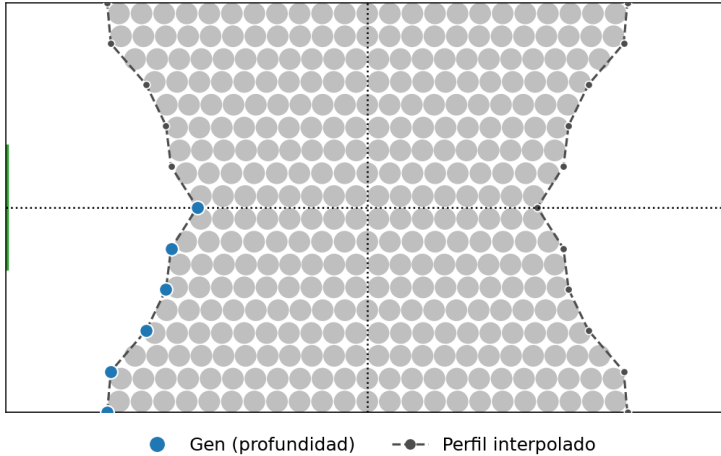
$K = 9$



Quad: mejor resultado

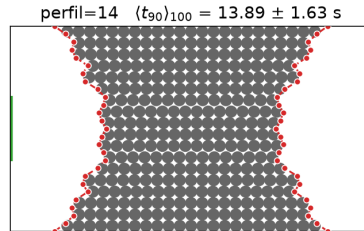
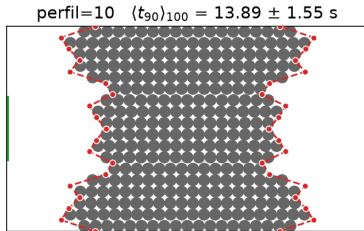
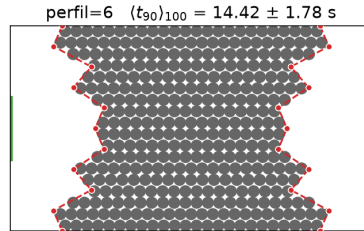
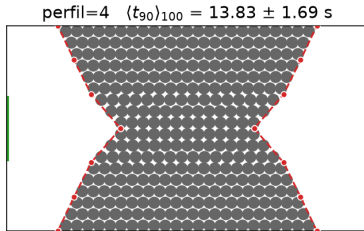


Video: <https://youtu.be/TODO>

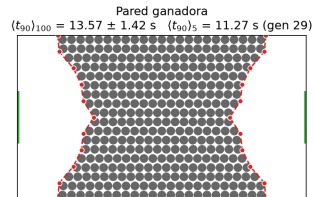
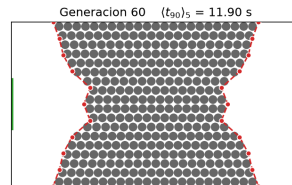


Pared: geometrías por puntos de perfil

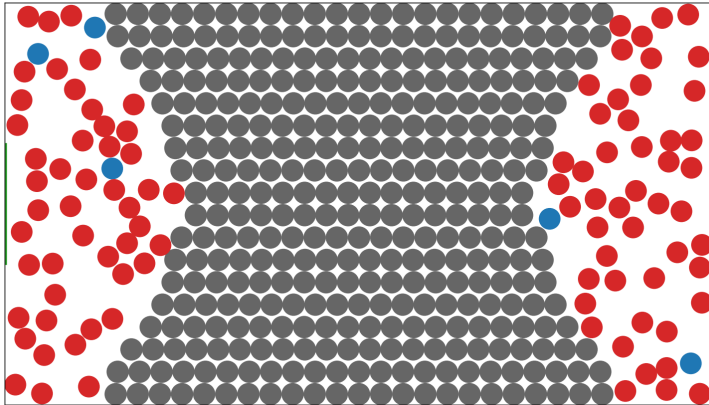
Grano 0.018 m, pobl. 40, 80 gen.



6 puntos, pobl. 64, 150 gen.

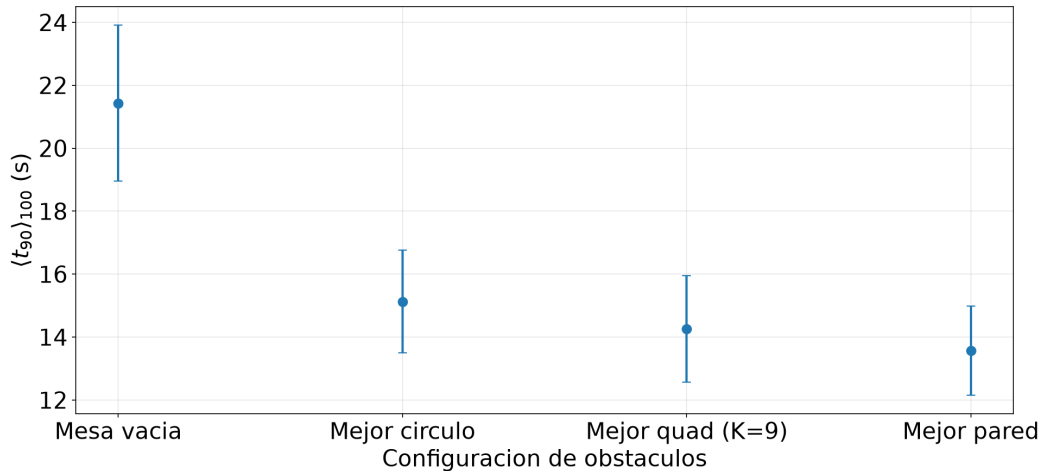


Pared: mejor resultado

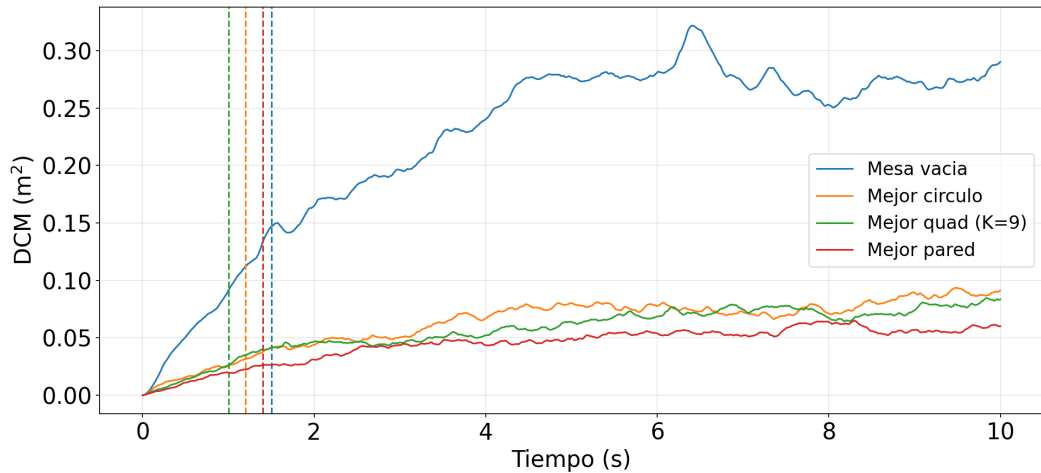


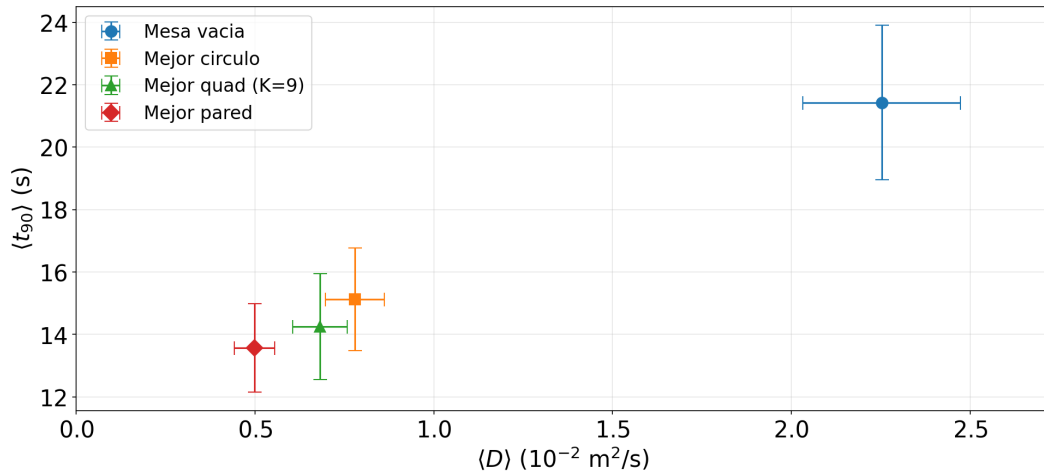
Video: <https://youtu.be/TODO>

Comparación final



DCM vs Tiempo



$\langle D \rangle$ vs $\langle t_{90} \rangle$ 

Conclusiones

Gracias!